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Outbreaks of European spruce bark beetle dramatically altered Norway spruce-dominated stands with implications for volant wildlife in the Białowieża Forest, Poland

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Abstract

Key message An outbreak of the European spruce bark beetle (*Ips typographus* L.) in spruce (*Picea abies* L.) dominated forest stands in the Białowieża Forest, Poland, has led to fundamental changes in forest characteristics. The species composition of regenerating stands now differs from earlier records findings, with oak, birch and rowan becoming dominant, while hornbeam numbers have decreased and Norway spruce is now sporadic, suggesting an ongoing period of reorganisation. Among the forest attributes showing the greatest degree of change are canopy architecture and sheltering opportunities—factors known to be most influential for volant-dependent wildlife.

Context The European spruce bark beetle (*Ips typographus* L.) targets Norway spruce (*Picea abies* (L.) H.). Once attacked, susceptible trees weaken and die. In planted forests with high proportions of Norway spruce, beetle outbreaks profoundly alter forest structure. The implications of such outbreaks on forest regeneration and their influence on dependent wildlife require ongoing research.

Aims We investigated how bark beetle-induced changes in Norway spruce stands alter forest structure characteristics and regeneration, and explored the potential consequences for forest-dependent volant vertebrates.

Methods We used a static comparative approach to assess differences in forest structure and regeneration between infested and non-infested Norway spruce stands in the Białowieża Forest, Poland. Eighteen forest variables were measured or derived, including indicators of canopy structure, tree composition, bark features, cavity presence, and microclimate. Sapling composition and regeneration frequency were also recorded to evaluate vegetation dynamics. Data were collected during 2022 and 2023.

Results Fifteen forest structural characteristics differed significantly post-infestation, including reduced tree density (trees per hectare), decreased vegetation clutter, and increased light transmittance. Daytime temperatures increased and humidity decreased, while nighttime conditions remained similar. The number of animal shelters in exfoliating bark increased, whereas tree crevices became less frequent. Sapling regeneration was higher in infested stands

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and dominated by oak (*Quercus* spp. L.), birch (*Betula pendula* Roth.), and rowan (*Sorbus aucuparia* L.). In contrast to earlier research, hornbeam was not the dominant sapling species, and Norway spruce saplings were infrequent.

Conclusion While our static comparative approach did not capture temporal dynamics or fully disentangle complex ecological interactions, it provided clear evidence that bark beetle outbreaks dramatically alter Norway spruce-dominated stands. The forest attributes most altered by infestation are well-established drivers of habitat suitability for forest-dwelling volant wildlife. We propose that cyclical beetle outbreaks in spruce-dominated forests act as drivers of accelerated natural succession, shifting species composition from planted Norway spruce to more naturally occurring species. Our findings highlight that bark beetle outbreaks represent natural change, not forest destruction, within this unique European forest ecosystem.

Keywords Pest infestation, Primeval forest, Canopy, Regeneration, Saplings, Structure

1 Introduction

Non-anthropogenic stochastic events have the potential to radically change the structure of forest habitat and reshape forest dynamics (Grebner et al. 2013). Large-scale disturbances, like windthrow or insect outbreaks, show a greater magnitude of impact in some years than others depending on climatic conditions and forestry management (Washaya et al. 2024; Selikhovkin et al. 2022).

The European spruce bark beetle *Ips typographus* (L) targets Norway spruce *Picea abies* (L) H. Karst as host trees (Wermelinger 2004). Adult male beetles bore into susceptible spruce trees (thinned barked low resin trees) while releasing pheromones to attract mates. Adults then copulate, and their larvae create tunnels in the phloem of the tree while feeding (Wermelinger 2004). The damage caused to the host tree often leads to death. Cyclical outbreaks of bark beetles can lead to increased ecological function as susceptible spruce trees are killed, reducing competition for resources from less susceptible trees (Selikhovkin et al. 2022; Müller et al. 2008). Recent outbreaks of bark beetle in the partly natural Białowieża Forest complex and planted stands in Leningrad province, St. Petersburg, are examples of large-scale impact, attributed to anthropogenic activity (i.e. planting of Norway spruce

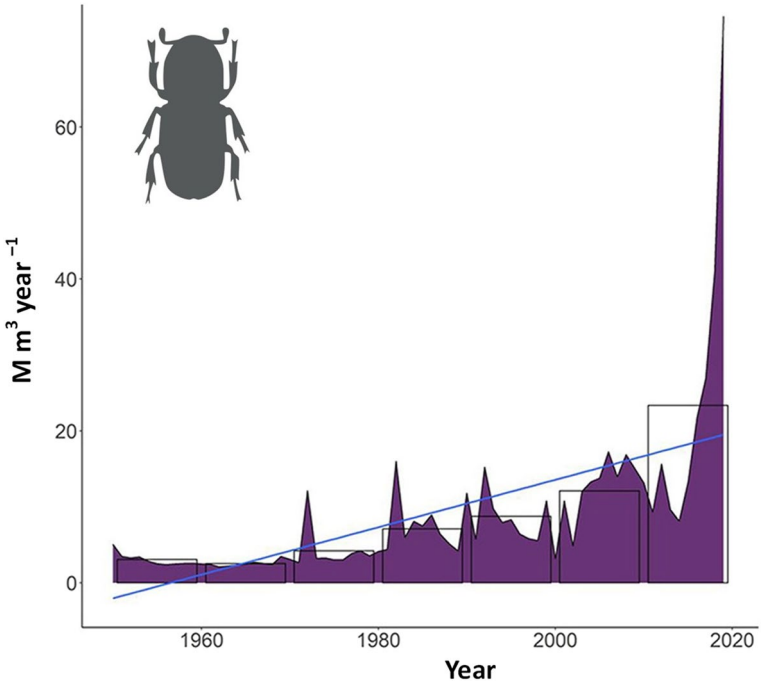

Fig. 1 Results of gap-filled time-series of bark beetle disturbance by Patacca et al. (2023) showing a large increase in bark beetle disturbance in the decade 2010–2019. The values represent the sum of the 34 European countries. The bars represent a decadal average. Lines are linear models fitted to the decadal averages

monoculture stands) compounded by favourable climatic conditions for beetle infestations (Kamińska et al. 2022; Selikhovkin et al. 2022) or stand-level attributes such as tree age (Kamińska et al. 2020). Bark beetle outbreaks in Central Europe have intensified during the last two decades (Washaya et al. 2024) with larger cyclical disturbances occurring after 2000 in line with hot and dry years, reaching an average of 23 M m^3 $year^{-1}$ in the decade 2010–2019 (Patacca et al. 2023) (Fig. 1). The consequence is economically negative and ecologically profound as whole stands of trees die and fall, relating more to clearcut forestry than natural forest gap dynamics. Although the negative impact on stock is clear, the future influence of forest tree species and composition, or the consequence on dependent wildlife, is not.

Despite uncertainty, significant structural changes can be expected to affect both tree regeneration (Dobrowolska et al. 2022a) and various animal groups, from insects (Busse et al. 2022) to flying vertebrates such as birds, for which hideaways and forest clearings are key for breeding (Küfner and Utschick 2016; Przepióra et al. 2020), while forest fragmentation has negative effects (Morante-Filho et al. 2018). Bats may benefit from new roosting opportunities (Rachwald et al. 2022b) or increased food resources (Divoll et al. 2022), and ungulates show varied responses—for example, *Cervus canadensis* Erx. and *Alces alces* L. increased their activity following bark beetle outbreaks, while *Odocoileus hemionus* Raf. responded negatively (Ivan et al. 2018), and *Bison bonasus* L. made greater use of beetle-damaged stands (Klich et al. 2023). Additionally, forest management that shapes structure can influence ungulate behaviour and reduce damage caused by deer (Reimoser et al. 2021), and in some cases improve physical condition (Reiner et al. 2023).

Disturbance is recognised as a key factor in the dynamics of forest vegetation (Seidl et al. 2011). Understanding the role of natural disturbance in long-term forest dynamics, including post-disturbance regeneration, is becoming increasingly important in the face of climate change (Dale et al. 2001). Predicting the impact of infestation on floral species regeneration is difficult due to complex biotic and abiotic interactions, although general assumptions are possible. Seidl and Turner (2022) propose that a short window post disturbance is pivotal; during this reorganisation period, the establishing species will shape forest structure and composition for centuries—in species-poor monocultures, reorganisation will result in changes of composition and structure. Altered canopy cover will change available light, influencing which plant species will likely regenerate (Dobrowolska et al. 2022a). Pioneering species will greatly influence future composition and determine the amount of light transmittance at the forest floor. Exploring plant regeneration and temporal changes in forest structure post infestation is a balanced approach for assessing broader impacts on forest ecosystem as a whole.

Here we quantified the change in forest characteristics such as canopy openness, number of trees and amount of light transmitted to the forest floor, approximately 12 years after the first signs of the largest recorded outbreak (Kamińska et al. 2022) of *I. typographus* in the Białowieża Forest, Poland. We explored how changes in forest characteristics affected sapling regeneration and tested two hypotheses: (1) bark beetle outbreaks in Norway spruce-dominated stands significantly alter forest structural characteristics, and (2) these structural changes influence the flight ability of bats and birds (number of standing trees and clutter) and increase the availability of high-quality shelters for forest-dependent volant wildlife. We considered likely future forest tree species composition and discussed the possible response of forest-dwelling animals.

2 Materials and methods

2.1 Study area and site selection

Białowieża Forest (52°45′29′′ N, 23°46′8′′ E) contains the last remaining areas of primeval forest in the Central European lowlands (Jaroszewicz et al. 2019). The forest complex borders Poland and Belarus and covers approximately 1500 km^2 (Samojlik et al. 2020). While sections of the forest are designated and afforded protection, on the Polish side, the contiguous forest includes diverse forest types including monospecific spruce stands. Coniferous stands are mostly a result of twentieth century planting, cover more than 25% of Białowieża Forest and consist of economically significant Norway spruce and Scots pine *Pinus sylvestris* (L.) (Stereńczak et al. 2022). Deciduous forests remain dominant with notable species including hornbeam *Carpinus betulus* (L.), oak (*Quercus robur* L.), linden (*Tilia cordata* Mill.), alder (*Alnus glutinosa* L.) and birch (*Betula pendula* Roth. Rachwald et al. 2022a). Białowieża Forest experienced cyclical bark beetle outbreaks. The most recent outbreak began in 2012 (Stereńczak et al. 2022) with subsequent dieback ongoing in 2019 (Kamińska et al. 2022). Some Norway spruce stands displayed high numbers of dead trees, while others remained unaffected (Stereńczak et al. 2022).

To assess the effects of bark beetle outbreaks, we employed a static comparative approach by measuring forest structure and regeneration in infested and non-infested Norway spruce stands. This space-for-time substitution allowed us to infer outbreak-driven changes without pre-outbreak data. We selected 15 Norway

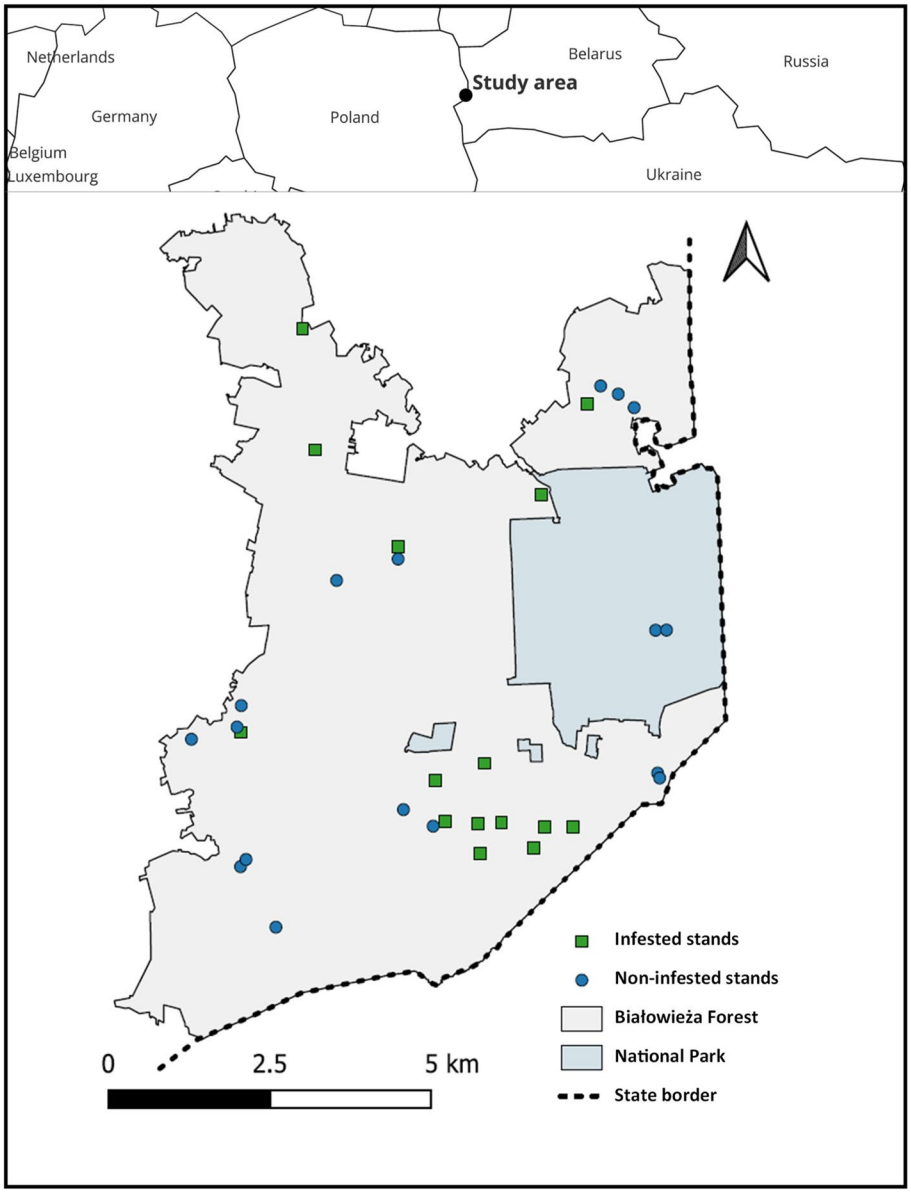

Fig. 2 Map of the Białowieża Forest showing the location of 15 infested and 17 non-infested Norway spruce dominated stands (sample plots). Each plot was surveyed for forest characteristics and regeneration between 2022 and 2023. Plots were distributed throughout the forest complex including locations within the strict reserve, nature reserves and non-protected areas

spruce-dominated stands that had been infested by *I. typographus*, and 17 ones that displayed no sign of past or current infestation (Fig. 2). Sites were selected through a combination of mapped infested and non-infested stands using remote sensing data collected by the Forest Research Institute as part of the LIFE + ForBioSensing project (Kamińska et al. 2022), and through field surveys, in three stages: firstly, stands dominated by spruce were selected using the Forest Data Bank (https://www.bdl.lasy.gov.pl/portal/mapy). This identified that spruce stands were unevenly distributed across Białowieża Forest, occupying approximately 25% of its area, ruling out the possibility of randomly distributing the study areas across the entire forest. Secondly, we selected stands transformed by bark beetle and those with no evidence of a beetle outbreak (living stands) using available data from the Forest Research Institute including remote sampling data showing transformed and living spruce stands that were at least 60 years old. Lastly, we visited each site to validate the categories and further assess their

characteristics using detailed criteria including preliminary counts of trees, tree species composition, and observations of characteristics that would make the stands unique. We observed that the data on tree stands from remote sensing were often inaccurate and the final selection of study points often had to be based mainly on the results of a site inspection, correcting the earlier selection. In practice, this often meant abandoning a planned location and choosing another one, possibly in close vicinity. Sites considered to be evenly mixed with Scots pine were not included. An additional site selection criterion was that there were no signs of management such as felling or dead tree removal, as was observed at sites adjacent to roads or forest tracks. It was also necessary to ensure that the distance from other open areas and from roads was sufficient to prevent these habitats from interfering with the results. A distance of at least 50 m was the cut-off. The final distribution of study points is a result of the spruce distribution in our study area, the spatial distribution of the bark beetle outbreak, and initial forestry interventions by the State Forests taken in the first stage of the outbreak (salvage logging and planting new trees). We consider our approach for site selection meant that any observed changes in the structural attributes between infested and non-infested stands of a study plot were a result of infestation, and that sites were similar in forest type and characteristics before infestation.

2.2 Survey of forest characteristics

To obtain comparable stand characteristic variables, one 20 × 20 m quadrat was identified in each plot (Carr et al. 2020). Measured or calculated characteristics included (i) tree species composition and stand structure, (ii) tree diameter at breast height and basal area per hectare, (iii) light transmittance, (iv) roosting/nesting opportunity (tree crevices and protruding bark), (v) air temperature and humidity and (vi) tree regeneration (see Table 1 for further detail of all variables) (Carr et al. 2025).

Each tree within a quadrat was (i) identified to species, (ii) classified as live or dead, (iii) surveyed for % of exfoliating bark and (iv) surveyed for cavities from the ground using × 10 binoculars. We considered exfoliating bark (bark plates) and tree cavities only if they were potentially suitable to be used as shelter for bats and birds (Goodall et al. 2019; Carr et al. 2020). Understory vegetation clutter was measured between 2 and 10 m in height from the ground at nine gridded points within each quadrat (e.g. using 5-m spacings along the edge of the quadrat every line intersection within each quadrat was a survey point. To ensure an appropriate level of accuracy a guide line with 5-m markings was placed on two sides of each plot). Adapting a method used by Smith and Gehrt (2010), a 10-m pole with 10 subsections marked on it was placed at each point. Any vegetation touching a subsection was counted and summed to provide a measure of clutter from 0 to 100%. Temperature and humidity were sampled using ibuttons (Maxim Integrated, US) placed on a tree at the centre of each plot facing away from direct sunlight. Canopy openness and transmittance of sunlight were measured at the same nine gridded points using a fish-eye lens connected to a smartphone positioned at head height and pointed upwards (Bianchi et al. 2017). Percent of canopy openness was determined through true colour hemispherical photography and Gap Light Analyzer software (Frazer et al. 2021). The fraction of incident light transmitted below the canopy (Comeau 2000) was determined. We counted the number of fallen trees in a subset of sites in each plot to determine if the infested and non-infested stands had similar numbers of trees post outbreak (Carr et al. 2025).

2.3 Survey of regeneration

Seedling and sapling regeneration was recorded in each of the 32 plots (Carr et al. 2025) using a standardised method (Dobrowolska et al. 2022b). From a central point of each quadrat all seedlings up to 0.5 m high within a 1.78-m-radius circular plot were counted and identified to species or genus. From the central point of each quadrat all saplings with a height > 0.5 m and DBH < 7 cm within a 5.64-m-radius circular plot were counted and identified to species or genus, and assessed for health condition. We used the Forest Data Bank (https://www.bdl.lasy.gov.pl/portal/mapy) online repository to extract additional covariates including forest site (habitat) type (e.g. dry, moderate ('fresh') or wet sites, conifer or broad-leaved and a combination of the two, i.e., conifer wet and conifer 'fresh'), following forest site type classification, developed on the basis of climatic, hydrological and soil characteristics (Matuszkiewicz 1978; Lasota and Błońska 2013). Dry sites were not present in the study area. The terms 'dry', 'fresh' or 'wet (bog)' are derived from the groundwater level.

2.4 Statistical analysis

To explore how the collected forest variables differed between infested and non-infested Norway spruce-dominated stands, we used a univariate regression approach using one predictor (x) variable (infestation status [infested/not infested]). All models were mixed, with site location included as a random variable to control for potential spatial autocorrelation. We used 5 km as the cut-off (i.e. all sites within 5 km of another site were grouped) resulting in a random variable with 12 groups. We performed a series of Generalized Linear Mixed Models (GLMMs) using the lme4 package (Bates et al. 2015) [model <-—glmer (response ~ predictor + (1

Table 1 Surveyed habitat variables recorded in the Białowieża Forest in September 2022 and 2023 including survey method

Habitat variable	Measurement
Bark plates	% exfoliating bark on all trees assessed as providing shelter for tree-dwelling animals. All trees within a quadrat were surveyed from the ground. Three separate surveyors observed a tree from three different sides, each providing an estimation of exfoliating bark. If two or more surveyors gave the same value the mode was used. If all three surveyors gave different estimates the median was used
Basal area m^2 per ha	Total standing wood including both live and dead trees. Basal area was measured using the DBH of all trees within a quadrat using the formula PI()*(DBH^2)/40000 and an expansion factor of 25
Canopy openness	% canopy openness determined though true colour hemispherical photography and Gap Light Analyzer software (Frazer et al. 2021). A fisheye lens connected to a smart phone positioned at head height and pointed upwards was used for photography (Bianchi et al. 2017). The phone was orientated to point north. The average was taken from nine gridded sampling points
Clutter	% clutter from two metres to 10 m from the ground. A 10-m pole with 10, 1-m subsections marked on it was placed at nine gridded points within each quadrat. Any vegetation touching a subsection was counted and summed to provide a measure of clutter from 0 to 100% (Smith and Gehrt 2010; Lintott et al. 2015; Carr et al. 2020)
DBH	Mean average of tree Diameter at Breast Height. The DBH of all standing trees within each quadrat were measured and the average value used
Humidity	Measure of relative humidity taken from humidity loggers (ibuttons; Maximum Integrated) placed within each site over a 24-h period, day (sunrise to sunset) and night (sunset to sunrise). One logger was used per site and placed on the most central tree at head height. All loggers were placed on the northern side of the tree to avoid direct sunlight, and left in situ for two months
Number of cavities per ha	Visual count of cavities from the ground using × 10 binoculars. Cavities included splits, cracks, woodpecker and rot holes suitable for tree-dwelling bats and birds, but did not include cavities under bark plates
Number of broadleaved trees per ha	Count of all broadleaved stems that were at least 2 m in height
Number of dead trees per ha	Count of all standing dead trees
Number of live trees per ha	Count of all stems that were at least 2 m in height and assessed as being alive
Number of trees per ha	Count of all stems that were at least 2 m in height
Number of tree species per Ha	Total number of tree species or genus
Number of trees with PRFs per ha	Potential Roost Feature (PRF). Total number of trees that had exfoliating bark and or at least one cavity assessed as providing shelter for tree-dwelling bats and birds
Sapling regeneration	Count of all saplings with a height > 0.5 m and DBH < 7 cm within a 5.64-m-radius plot from the centre of each quadrat
Seedling regeneration	Count of all seedlings up to 0.5 m in height within a 1.78 m radius plot from the centre of each quadrat
Transmittance	Fraction of incident light transmitted below the canopy (Comeau 2000) determined through true colour hemispherical photography and Gap Light Analyzer software (Frazer et al. 2021). A fisheye lens connected to a smart phone positioned at head height and pointed upwards was used for photography (Bianchi et al. 2017). The phone was orientated to point north. The average was taken from nine sampling points within each quadrat
Temperature	Measure of temperature (°C) taken from temperature loggers (ibuttons; Maximum Integrated) placed within each site over a 24-h period, day (sunrise to sunset) and night (sunset to sunrise). One logger was used for each quadrat and placed on the most central tree at head height. All loggers were placed on the northern side of the tree to avoid direct sunlight and left in situ for two months

Unless otherwise stated, all measurements were taken within 20 × 20 m quadrats and extrapolated to per hectare where appropriate using an expansion factor of 0.25 (400 m^2 × 25 = 1 ha)

| random), family = "", data = data]. Model residuals were visually assessed to select the most appropriate distribution from Gaussian, Poisson or negative binomial. An adjusted R^2 ('sjPlot' package (Lüdecke 2024)) was used to consider the amount of variance explained by the single binary predictor variable. Data were not standardised, and the model structure of one binary predictor variable means the effect size estimates are actual observed values regardless of which error distribution was used. Outliers were assessed visually during data exploration by looking at the raw data, using scatterplots and assessing the spread of data in box and whisker plots. None were considered influential. We tested whether pooled standing and fallen trees were significantly different between treatments (infested and non-infested sites) using a Wilcoxon test. All analyses were performed within R v.4.1.2 (R Core Team 2021).

To determine which forest structure characteristics most influenced sapling regeneration abundance (or density), we built a series of GLMMs for the four regeneration response variables most affected by bark beetle infestation [model <—glmer (response ~ $predictor^1$ + $predictor^2$… + (1 | random), family = "", data = data]. These response variables included all saplings (species

combined), oak (*Quercus* spp.), birch (*Betula* spp.) and rowan (*Sorbus aucuparia*) saplings. Of the total measured forest characteristics, only those assessed as meaningful were considered for use as predictor variables. For example, types of variables such as canopy openness and transmittance of light that likely influence the regeneration of seedlings and saplings were considered. Variables that describe animal roosting or nesting potential such as the number of tree cavities were not considered. Multicollinearity between the selected predictor variables ($n=17$) was assessed prior to model building using Spearman's rho correlation coefficient and a $|r|>0.7$ coefficient threshold (Carr et al. 2025). Those above 0.7 were removed to avoid distorting model estimation and predictions (Dormann et al. 2013). Canopy openness, transmittance, temperature and humidity (we collected temperature and humidity using ibuttons continuously and divided these data into 24-h periods, day and night. We considered all periods potentially biologically meaningful but given 24-h periods were correlated (>0.7) with either day or night periods the simplest approach was to remove 24-h period data. This also meant we could retain the differences during the day and night which we subsequently found to be informative) were highly correlated with each other. We removed multi-correlated variables while also considering the relevance and usefulness of the remaining covariates. The subsequent number of predictor variables was eight, including canopy openness, daytime temperature, daytime humidity, basal area, nighttime humidity, nighttime temperature, stand type and water level (see Sect. 2.3 for details of stand categories). We consider these biologically meaningful measures of the forest environment that are key drivers of regeneration (Dobrowolska et al. 2022b). The error distribution was selected by visually assessing the distribution of model residuals. To identify the most parsimonious model that also explained an acceptable amount of variance, Akaike's Information Criterion adjusted for small sample sizes (AICc) was used ('AICCmodavg'; Mazerolle 2023). We used delta AICc (ΔI) ≤ 2 and a step-down approach to remove variables resulting in a final model with only variables from the best fitting models (Burnham and Anderson 2002). The responses of significant variables were plotted using the 'sjPlot' package (Lüdecke et al. 2023). Outliers were assessed by assessing residuals and against fitted value plots. A single data point for humidity was removed as it was found to influence the results.

Infestation status (infested/not infested) was included in multicollinearity analysis and found to be highly correlated with a large proportion of covariates (Carr et al. 2025). Our method of separation (univariate regression followed by GLMMs) reflects the dominant influence of infestation status on covariates. To assess whether spatial autocorrelation remained in the residuals of our models, we conducted a Moran's *I* test to evaluate the degree to which similar values are clustered in space, with Moran's *I* values ranging from −1 (strong dispersion) to +1 (strong clustering), and values near zero indicating spatial randomness. Residuals from each model were extracted and used in the spatial autocorrelation analysis. To construct the spatial weights matrix, we used the four nearest neighbours ($k=4$) based on the geographic coordinates (latitude and longitude) of each sampling location, employing the knearneigh() and nb2listw() functions from the spdep package. A row-standardised weighting scheme ('W' style) was applied. We then performed the Moran's *I* test on the residuals using the moran.test() function, with permutation-based significance testing and the zero.policy = TRUE option to handle locations with no neighbours.

To explore the effect of time since bark beetle infestation on forest regeneration, we used Spearman's rank correlation tests. The response variables included sapling counts of four tree species (birch, Norway spruce, oak, and rowan), total saplings, total seedlings and species richness (defined as the number of species). Time since infection was established using historical aerial imagery taken from Google Earth. When signs of infestation were visible in a satellite image, that year was used.

3 Results

3.1 Forest structure

The number of pooled standing and fallen trees in infested (675 ± 200 IQR) and non-infested (770 ± IQR [Interquartile range]) stands were not significantly different ($W=30.5$, $p=0.3669$). We found 14 of the 17 measured forest characteristics were significantly different in infested Norway spruce stands when compared to non-infested stands (Table 2; Carr et al. 2025). Those characteristics considered to be influential to sapling regeneration (See Sect. 3.2) and volant forest—and tree-dwelling faunal species showed the largest differences. Our models predicted 400 fewer trees (± 71 SE) ($p<0.001$), reduced basal area (844 ± 153 SE) ($p<0.001$) and clutter (3.25 ± 1.15% SE) ($p<0.01$) in infested stands. The large reduction in trees was mostly due to fewer Norway spruce, although broadleaved trees also reduced by 100 (± 42 SE) ($p=0.03$) per ha. The number of trees categorised as 'live' significantly reduced (488 ± 55 SE) ($p<0.001$), and the number of standing dead trees significantly increased by 90 (± 31) ($p<0.01$) per ha.

The number of potential roosting and nesting features (trees with potential roost features (PRFs)) was significantly greater in infested stands (115 ± 27 ha^{-1} SE). We further categorised PRFs as exfoliating bark

Table 2 Results of univariate generalised linear models using habitat variables recorded from *Ips typographus* infested ($n = 15$) and non-infested Norway spruce dominated stands ($n = 17$) within Białowieża Forest (refer to Table 1 for description of each variable). Each model included one binary predictor variable (infestation status). Each model used an appropriate distribution. The list of independent variables is ordered from highest to lowest *z*-/*t*-value. *P*-values are given as *$p < 0.05$ **$p < 0.01$ ***$p < 0.001$. *ns* not significant. The distribution used is indicated in the response variable column as *g* Gaussian, *p* Poisson, *b* binomial

Independent variable	Predictor variable	Estimate	SE	z/t-value	*p*	Adj. R-squared
Transmittance g	Infestation status	51.46	4.22	12.20	***	0.83
Canopy openness (%) g		43.98	3.75	11.71	***	0.81
No. live trees (ha^{-1}) p		−487.94	55.01	−8.87	***	0.71
Bark plates (%) p		17.58	2.40	7.32	***	0.63
Daytime temperature (°C) g		1.13	0.18	6.39	***	0.56
Basal area (m^2 ha^{-1}) g		−21.67	3.66	−5.93	***	0.53
No. trees (ha^{-1}) p		−400.59	71.16	−5.63	***	0.50
24 h temperature (°C) g		0.82	0.16	5.00	***	0.44
Daytime humidity g		−6.17	1.27	−4.88	***	0.42
No. trees with PRFs (ha^{-1}) p		115.10	26.27	4.38	***	0.37
24 h humidity (°C) g		3.98	1.23	3.22	**	0.23
No. dead trees (ha) b		90.10	30.93	2.91	**	0.19
Clutter (%) g		−3.25	1.15	−2.82	**	0.25
No. cavities (ha^{-1}) b		−117.45	47.53	−2.47	*	0.13
No. broadleaved trees (ha^{-1}) p		−100.49	42.63	−2.36	*	0.13
Nightly temperature (°C) g		−0.43	0.22	−1.95	ns	0.08
DBH (m) g		1.36	1.71	0.79	ns	0.01
Nightime humidity g		0.54	1.15	0.47	ns	0.03

(bark plates) and tree crevices (Table 1). In line with a significant increase of standing dead trees ($p < 0.01$), bark plates assessed as providing roosting and nesting opportunity significantly increased (18 ± 2% SE) ($p < 0.001$) following infestation (Fig. 3). In contrast, the number of tree cavities assessed as providing roosting and nesting opportunity significantly decreased by 117 in infested stands (± 48 ha^{-1}SE) ($p < 0.05$).

Canopy openness was 44% (± 4% SE) ($p < 0.001$) larger in infested stands due to the reduction in basal area and number of trees and loss of needles (Fig. 4) as a result of infestation. The amount of transmittance through the canopy (transmittance) was significantly greater (51 ± 4% SE) in infested stands with similarly high estimates and relatively small standard errors (Table 2; Carr et al 2025). Adjusted R-squared values were greatest for canopy openness ($R^2 = 0.83$) and transmittance ($R^2 = 0.81$) followed by the number of live trees ($R^2 = 0.71$) and bark plates ($R^2 = 0.63$) (Table 2).

Temperature during the day was 1.13 °C higher in infested stands (± 0.2 °C SE) ($p < 0.001$). In contrast, relative air humidity during the day was significantly lower (6.17 ± 1.27 SE) ($p < 0.001$) (Table 2; Carr et al 2025). Temperature and humidity levels during the night were similar between infested and non-infested stands (Table 2), as were the number of tree species and average DBH.

3.2 Forest regeneration

Sixteen species of saplings were documented with clear differences in presence and abundance between infested and non-infested stands; the most frequent species included birch, oak, rowan and spruce (Carr et al. 2025). We found four measured regeneration variables to significantly change in infested Norway spruce stands when compared to non-infested stands (Table 3). The number of saplings among species within non-infested stands was significantly different, with more spruce saplings

Fig. 3 Visual examples of structural characteristics in infested and non-infested Norway spruce stands in the Białowieża Forest, Poland. Top left: hemispherical photo taken in a non-infested stand with a sampled site average canopy openness (29%) and transmittance (28). Top right: hemispherical photo taken in an infested stand with a sampled site average canopy openness (79%) and transmittance (67%). Middle left: example of a non-infested Norway spruce dominated stand. Middle right: example of an infested Norway spruce dominated stand. Bottom left: example of a non-infested Norway spruce tree which typically does not have protruding bark that could be considered suitable for tree-dwelling animals. Bottom right: example of a Norway spruce tree following infestation by *I. typographus.* Post infestation, the tree weakened and died, and the bark began to protrude from the trunk providing shelter for tree-dwelling animals

($p < 0.05$) than oak and birch saplings. The number of saplings was greater ($p < 0.01$) in infested stands, with 34 (± 10 SE) more saplings compared to non-infested stands. The condition of saplings and the number of seedlings were not significantly different between infested and non-infested stands. When subdividing saplings into genus, the number of oak, birch and rowan saplings was significantly greater in infested stands (Table 3) with pioneering birch saplings having, on average, 24 (± 9 SE) ($p < 0.01$) more individuals in infested stands. Oak had on average three (± 1 SE) and rowan eight (± 3 SE) more saplings per plot in infested stands when compared to non-infested stands. We found no significant relationships between time since infestation and any regeneration variable. With the exception of one site, all sites showed signs of infestation within 3 years of each other.

We further explored which forest characteristics influenced the frequency of saplings by selecting forest characteristics we considered relevant (see

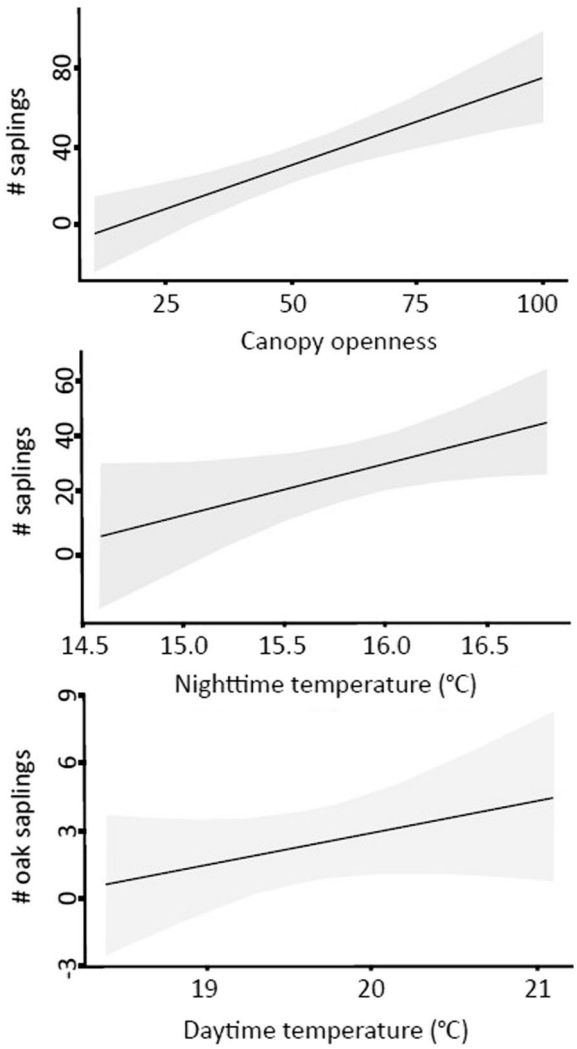

Fig. 4 Predicted response of total number of saplings and oak saplings to significant predictor variables present in the best models (ΔI ≤ 2) using a generalised linear mixed model with a Poisson error distribution for saplings and binomial for oak saplings. Sapling frequency was positively influenced by canopy openness ($p < 0.001$) and temperature during the night ($p < 0.05$). Oak saplings were positively influenced by temperature during the day ($p < 0.01$)

Sect. 2.3). The best model for birch saplings indicated a negative influence of canopy openness and positive influence of humidity levels during the day. Stands categorised as wet were predicted to have a greater number of birch saplings than 'fresh' (i.e., of moderate humidity (Matuszkiewicz 1978)) stands. Although none of the predictor variables were significant, the model explained a relatively large amount of variation (61%; adjusted $R^2 = 0.61$). Oak saplings responded positively to day temperatures ($p > 0.01$) (Fig. 4) and were influenced by stand category; most importantly that more oak saplings were present in wet mixed coniferous stands. The model with combined sapling numbers showed a positive response to canopy openness ($p < 0.001$) and nighttime temperatures ($p < 0.05$) (Table 4; Fig. 4).

Table 3 Results of univariate generalised linear models using regeneration variables recorded from *Ips typographus* infested ($n = 15$) and non-infested Norway spruce stands ($n = 17$) within Białowieża Forest. Each model included one independent variable and one binary predictor variable (infestation status). Each model used an appropriate distribution including Gaussian, Poisson and negative binomial. The list of independent variables is ordered from highest to lowest *z*-/*t*-value. *P*-values are given as *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. *ns* not significant. The distribution used is indicated in the response variable column as *g* = Gaussian, *p* = Poisson, *b* = binomial/logistic

Independent variable	Predictor variable	Estimate	SE	*z*-/*t*-value	*p*
Saplings p	Infestation status	33.8	9.55	3.54	**
Oak saplings b		2.55	0.86	2.96	**
Birch saplings p		24.3	8.77	2.78	**
Rowan saplings b		7.82	3.17	2.45	*
Spruce saplings b		−0.33	0.21	−1.05	ns
Sapling condition g		0.06	0.20	0.29	ns
Seedlings p		−1.32	1.27	−1.04	ns

4 Discussion

The most recent cyclical outbreak of bark beetle within the Białowieża Forest (Stereńczak et al. 2022) has resulted in significant changes in structural characteristics of Norway spruce dominated stands. Twelve years after the first recorded indications of an outbreak, we found that 14 of 17 measured forest characteristics were significantly altered (Table 2). Although the current outbreak of bark beetle within the Białowieża Forest is considered one of the largest recorded (Stereńczak et al. 2022), it is reasonable to suggest that any size outbreak of bark beetle will alter forest dynamics (Grebner et al. 2013). Although the resulting influence of bark beetle on Norway spruce forests is well studied (Schroeder and Lindelöw 2002; Wermelinger 2004; Müller et al. 2008; Hlásny et al. 2019; Kamińska et al. 2022; Selikhovkin et al. 2022; Stereńczak et al. 2022), our results are suitable to quantify change and measure elements of forest regeneration within a specific time period. As dynamic systems, temporal change following disturbances within forests is unpredictable and the drivers of change are difficult to identify when most forest characteristics are interrelated (Grebner et al. 2013). Before-After Control-Impact research is more suited to disentangling changes over

Table 4 Results of generalised linear models using regeneration variables recorded from *Ips typographus* infested ($n = 15$) and non-infested Norway spruce stands ($n = 17$) within Białowieża Forest (refer to Table 1 for description of variable collection method). Each model included one response variable and several predictor variables. Model selection followed a bottom-up approach using AICc (ΔI) scores. If the model reduced by $\Delta I \leq 2$ when a variable was included, that variable was not used in the final model. McFadden adjusted R-squared was also used at each inclusion of a covariate to assess the amount of variation explained. The method of finding the simplest model that explained the most variance was followed. The rowan sapling model was no better than the null model. Con.wet = wet mixed coniferous, Con.fresh = 'fresh' mixed coniferous, Broad.fresh = 'fresh' broadleaved and 'fresh' mixed broadleaved, Broad.wet = wet broadleaved, bog mixed broadleaved and wet mixed broadleaved. Each model used an appropriate distribution including Gaussian, Poisson and negative binomial. *P*-values are given as $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$. *ns* not significant. The distribution used is indicated in the response variable column as *g* Gaussian, *p* Poisson, *b* binomial

Response variable	Predictor variables	Estimate	SE	*t*/*z*-value	*p*	Adj- R-squared
Birch saplings b	Canopy openness	−0.07	0.03	−1.85	0.06	0.61
	Day humidity	0.49	0.28	1.79	0.07	
	Wet (Fresh)	3.73	2.15	1.73	0.08	
Oak saplings b	Con.wet – (Con.fresh)	3.09	1.81	1.71	0.09	0.31
	Broad.fresh – (Con.fresh)	1.33	1.20	1.11	0.27	
	Broad.wet – (Con.fresh)	−1.01	1.39	−0.73	0.47	
	Day temperature	2.31	0.87	2.66	**	
Saplings p	Canopy openness	0.91	0.21	4.32	***	0.45
	Night temperature	16.8	8.14	1.621	0.17	

time following stochastic events in complex habitats, but this is often not possible as before-event data were not collected, as was the case in our study. We evaluated temporal changes in regeneration across our infested stands but did not find any significant influence of time. This is most likely due to all of our infested sites—except for one—being infested within a three-year period. We consider that a longer interval between infestations would likely have resulted in significant differences in the number and composition of saplings and seedlings.

Spruce bark beetle is associated with Norway spruce (Kamińska et al. 2020). This explains the large decrease in total standing trees and an increase in standing dead trees that consisted mostly of spruce (Table 2). It also highlights that a decade after the start of the outbreak, a large amount of infested spruce trees had already fallen, indicating the temporal nature of change. This may have had some influence on our results; although, due to the surveillance of bark beetle signs within the study area, we can be sure that infestations started in each sample site within a 2-year period starting in 2012. The number of broadleaved trees was also less in infested stands. Although broadleaved trees are not directly impacted by spruce bark beetle, our results show they are indirectly impacted, potentially through damage by nearby falling spruce (Grebner et al. 2013) or higher susceptibility to blowdown as neighbouring infested spruce trees density is reduced (Alexander 1964).

Broadleaved trees included birch which is a common pioneer species in the European lowlands and occurs throughout the Białowieża Forest, both as young and mature trees (depending on the type and condition of the stand). Its presence in bark beetle sites results from its pioneer character. The same is true for the presence of rowan (another pioneer species in these conditions) in these sites. At the same time, the regeneration of the forest observed by deciduous species may indicate not only a return to the natural proportions of species (from before the forest management that preferred coniferous trees) but also the response of the Białowieża Forest ecosystem to climate change, preferring vegetation that is more climatically adapted (Bolte et al. 2014). Although Białowieża Forest is historically a forest with a predominance of deciduous trees, spruce is also a native species for this forest probably since the early Holocene (Dering et al. 2012).

We also observed a significant increase in canopy openness due to the reduction in the number of trees and loss of the assimilation apparatus in the remaining standing dead trees. The increased canopy opening then implies the higher daytime temperatures we found through increased sunlight (transmittance). This also resulted in the observed changes in humidity during the day, which reduced to a level below significance at night (Table 2).

An indirect result of the analysed changes is the occurrence of natural regeneration of the forest transformed by the bark beetle outbreak, which is the next stage of change taking place in tree stands after the outbreak.

We did not find any difference in seedling density between infested vs. non-infested forests. However, we observed that the numbers of oak, birch, and rowan saplings were significantly greater in infested stands, with birch saplings displaying the largest difference compared to oak and rowan. The increased canopy openings resulted in increased irradiance for regeneration, especially for light demanding tree species (Dobrowolska et al. 2022b). The lack of differences in the case of seedlings relates to the advance process of regeneration 12 years after the first recorded indications of the outbreak. There was no difference in seedlings and saplings growing in deciduous and coniferous habitat types; in both cases, the regeneration is similar and the typical regeneration for the first stage of succession, i.e. deciduous pioneer species dominate (Hynynen et al. 2010). Spruce-birch (*Betula* spp.) is the most common mixture on fertile mineral soils in Fennoscandia, as both species have similar site requirements and birch naturally regenerates well in juvenile spruce dominated stands, making it a tempting alternative to pure spruce stands (Männistö et al. 2024). *B. pendula*, *B. pubescens* E., *Populus tremula* L., *Salix caprea* L. and *Sorbus aucuparia* L. are pioneer tree species in hemi-boreal forests (Götmark et al. 2005) which benefit from disturbances and establish mainly in canopy gaps (Żywiec et al. 2013). For birch saplings, the model indicated a negative influence of canopy openness and a positive influence of relative humidity levels during the day. As birch is a light demanding pioneer tree species, it occupies easily open areas. Dobrowolska et al. (2022b) also found that birch regeneration was most likely in large gaps surrounded by spruces. Birch demands more moisture during the day because it can create longer and thinner short roots, ; a morphological adaptation leading to an increase water absorption capacity (Parts et al. 2013).

The dominance of deciduous regeneration is a manifestation of the first stage of natural succession (birch and rowan are pioneer species) but may also result from the fact that although Białowieża Forest is a mosaic of habitats, it is dominated by forest sites of deciduous type (Jaroszewicz et al. 2019). At the same time, it should be noted that even in potential deciduous forest sites, we recorded a low share of hornbeam in the regeneration (Carr et al. 2025). This contradicts an earlier opinion about the dominant role of this species in natural regeneration in Białowieża Forest (Brzeziecki et al. 2018; Dobrowolska et al. 2022b). Brzeziecki et al. (2018) observed an increased number of hornbeams, especially in the coniferous stands, which we did not observe in our research. We noticed successful regeneration of oak in post-disturbed areas. For oak saplings, we found a positive response to increased light (Kanjevac et al. 2021). Habitat type also influenced oak sapling presence. Better conditions for oak saplings were in wet mixed coniferous stands. Bobiec et al. (2011) also found more oak regeneration in mixed coniferous forest habitats in Białowieża Forest.

The model with combined sapling numbers showed a positive influence of canopy openness and nighttime temperatures. All tree species regenerating in the study area (birch, rowan, oak) are light demanding. Only spruce belongs to shade-tolerant tree species, but its regeneration was rare. All of them could easily occupy the post-disturbance spruce stands because of the favourable light conditions. Sites receiving greater solar radiation on average each day may have been warmer at night (Hill et al. 2019). Greater amounts of light may also have simply increased photosynthesis (Johnson et al. 2004).

Apart from the expected effect of creating space for natural regeneration of the forest, observed changes in forest characteristics and dynamics, for example, basal area, canopy architecture, light and temperature, humidity and the opportunity to shelter, all have the potential to directly and indirectly influence forest-dwelling species at different scales, impacting selection from general areas to individual sheltering sites. In addition to tree-dwelling vertebrates like birds and bats, the described factors also influence other animals, such as insects (Busse et al. 2022), as well as large animals, such as ungulates (Kujiper et al. 2009; Ivan et al. 2018).

Open areas in dense tree stands play an important role in the functioning of the ecosystem, including animal communities. Forest gaps provide feeding places for flying vertebrates (both birds and bats) (Fuller 2000; Przepióra et al. 2020; Latif et al. 2020; Erasmy et al. 2021; Divoll et al. 2022). Canopy gaps increase the diversity of the spatial structure and create an ecotone zone, a typical place of activity for birds and bats. In Białowieża Forest, the ecotone is an area of increased bat activity (Rachwald 1992), and changes in the structure of tree stands caused by the beetle have impacted bird populations (e.g. Wesołowski et al. 2015; Czeszczewik and Walankiewicz 2016). Tree-dwelling bats and birds are known to select different types of tree features (Suorsa et al. 2005; Bartonička et al. 2008; Carr et al. 2018; 2022; Goodall et al. 2019; Apoznański et al. 2021; Remeš et al. 2022). The relative density of barbastelle bats *Barbastella barbastellus* (Schr.) increased during the period following infestation (Rachwald et al. 2022a), and frequent use by barbastelles of dead spruce trees for maternity roosts has been observed (Rachwald et al. 2022b). Except for that, the increase in forest floor plant diversity, changes in the understory vegetation (Matuszkiewicz et al. 2024) and natural regeneration of tree species generally related

to increased exposure to light have an impact on terrestrial herbivores, an example of which is the reaction of European bison *B. bonasus* (Jaroszewicz et al. 2020). The importance of dead standing trees specifically in the Białowieża Forest for birds, as a food source for such groups like woodpeckers and nuthatches, and as nesting places, has been reported in several studies (Tomiałojć et al. 1984; Wesołowski 1989; Wesołowski and Rowiński 2004; Czeszczewik 2009; Czeszczewik et al. 2012; Walankiewicz et al. 2014). In the study area, we should expect all these processes to occur with considerable intensity, due to the naturally diversified structure of the forest and the significant size of the bark beetle outbreak in the Białowieża Forest.

More than a decade after the first signs of cyclical European bark beetle infestation in Norway spruce-dominated forest stands (and approximately 6 years after its end), we found most forest characteristics were still significantly different compared to non-infested stands. Many characteristics were influenced by each other, and, as hypothesised, those known to be influential for natural forest regeneration and dependent wildlife had the largest degree of change. The ecosystem of the Białowieża Forest, despite some changes caused by previous forest management, is still a largely dynamic and highly diversified natural system (Jaroszewicz et al. 2019). Norway spruce remains an important species in the Białowieża Forest, but occurs now in a more dispersed form in mixed tree stands. The changes we have identified are part of the dynamics of this unique contemporary European forest habitat. This cyclical bark beetle outbreak should not be described as a destruction of the forest (Goździewska 2016; Brzeziecki et al. 2018), but rather as a period of accelerated natural change (Stokstad 2017; Blicharska and Smithers 2018; Jaroszewicz et al. 2019).

5 Conclusion

Beetle outbreaks fundamentally alter Norway spruce-dominated stands. We found that the composition of saplings within infested stands had shifted from hornbeam to pioneer species, indicating a period of reorganisation, in contrast to data collected in 2012. Although our use of a static comparative framework limits inference about temporal dynamics, the consistent and substantial differences observed between infested and non-infested stands demonstrate the ecological significance of bark beetle outbreaks. Although disentangling complex forest interactions is beyond the scope of this paper, we propose that the observed bark beetle outbreak may ultimately benefit the forest, as natural regeneration will lead to structurally diverse, species-rich stands compared to the pre-outbreak monocultures. Although volant forest- and tree-dwelling wildlife likely benefited from the infestation during our sampling period, it is likely that over time, key forest characteristics influencing bats and birds—such as shelter—will decline (due to bark loss and falling trees) before increasing again in the long term as new trees mature. We propose that cyclical bark beetle outbreaks in spruce-dominated forests act as drivers of accelerated natural succession, shifting species composition from planted monocultures toward more naturally occurring assemblages. The structural changes we observed reflect a phase of accelerated natural dynamics, reinforcing that bark beetle outbreaks are not forest destruction, but part of the ecological cycle in this distinctive European forest. Post-outbreak, minimal-intervention forest management should be prescribed to Norway spruce stands to foster long-term resilience through increased species and age diversity. This approach will ultimately benefit forest-dependent wildlife, including protected species, in the face of ongoing global warming.

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Authors' contributions
AC, AR and GA conceptualised the research, inputted into methodology and investigated. AC undertook the formal analysis and visualisation and prepared the original manuscript draft. DD investigated. All authors were involved in manuscript review and editing. AR acquired funding and was responsible for project administration with assistance from GA.

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